

COMPSCI 1DM3 - Winter 2026

Discrete Mathematics for Computer Science

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Last Lecture:

- Proof is a valid argument that uses hypothesis and previously known facts to establish:

$$\forall x(P(x) \rightarrow Q(x)), \quad \text{for some domain}$$

or equivalently

$$P(c) \rightarrow Q(c), \quad \text{for any arbitrary } c \text{ in domain}$$

- Different proof techniques
 - ▶ Direct proof
 - ▶ Proof by contraposition
 - ▶ Proof by contradiction
 - ▶ Proof by cases

Chapter 2

Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

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- Informally, sets are used to group objects that have a similar property, e.g.,
 - ▶ The collection of all students enrolled in CS 1DM3
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- Sets are represented by curly brackets
- By convention, an empty group of objects is also considered to be a set; called **empty set** $\emptyset$

Sets

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- A is the same as $\{1, 5, 3, 9, 7\}$ and $\{1, 3, 5, 7, 7, 7, 9, 9, 9, 9\}$
- We say that 1 is a member of A , it is in A , or it belongs to A and we denote it by

$$a \in A$$

- Example: $13 \notin A$

Sets

How to describe sets:

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- Roster method: We write down all members of the set. For instance, the set of even numbers less than 10: $\{2, 4, 6, 8\}$
- Ellipses ($\dots$) method: If the set members have a nice property, then we write down only a few:

$$\{2, 4, 6, 8, \dots, 46, 48\}$$

- Set builder: Instead of writing down all members, we precisely describe the common property:
 - ▶ $A = \{x \in \mathbb{N} \mid x \leq 20\}$

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 - ▶ $C = \{x \in \mathbb{R}^+ \mid x \leq 20\}$
 - ▶ $D = \{x \in \mathbb{Z} \mid |x - 5| \leq 3\}$

Some notation

- Natural numbers: $\mathbb{N} = \{0, 1, 2, 3, 4, \dots\}$
- Integer numbers: $\mathbb{Z} = \{\dots, -4, -3, -2, -1, 0, 1, 2, 3, 4, \dots\}$
- Positive integer numbers: $\mathbb{Z}^+ = \{1, 2, 3, 4, \dots\}$
- Rational numbers: $\mathbb{Q} = \{\frac{p}{q} \mid p \in \mathbb{Z}, q \in \mathbb{Z}, \text{ and } q \neq 0\}$
- Real numbers: $\mathbb{R}$
- Positive real numbers: $\mathbb{R}^+$

Some notation

Let a and b be some real numbers.

- $[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$
- $[a, b) = \{x \in \mathbb{R} \mid a \leq x < b\}$
- $(a, b] = \{x \in \mathbb{R} \mid a < x \leq b\}$
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Sets

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- We say A and B are equal, $A = B$, if they have the same elements:
In other words, we have

$$A = B \quad \text{if and only if} \quad \forall x (x \in A \leftrightarrow x \in B)$$

Example:

- $\{1, 3, 5, 7\} = \{1, 1, 1, 5, 5, 7, 7, 3\}$? ✓
- $\{1, 2, 3\} = \{\underbrace{\{1, 2\}}, \underbrace{3}\}$? ✗

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- We say set A is a **subset** of B , $A \subseteq B$, if all elements of A are included in B :
In other words, we have

$$A \subseteq B \quad \text{if and only if} \quad \forall x(x \in A \rightarrow x \in B)$$

Example:

- $\{1, 3, 5, 7\} \subseteq \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$? ✓
- $\{1\} \subseteq \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$? ✓
- $\{\{1, 2\}\} \subseteq \{\{1, 2\}, 3\}$? ✓
- $\{1, 3\} \subseteq \{\{1, 2\}, 3\}$? ✗
- $\emptyset \in \{1, 2, 3, 4, 5\}$? ✓

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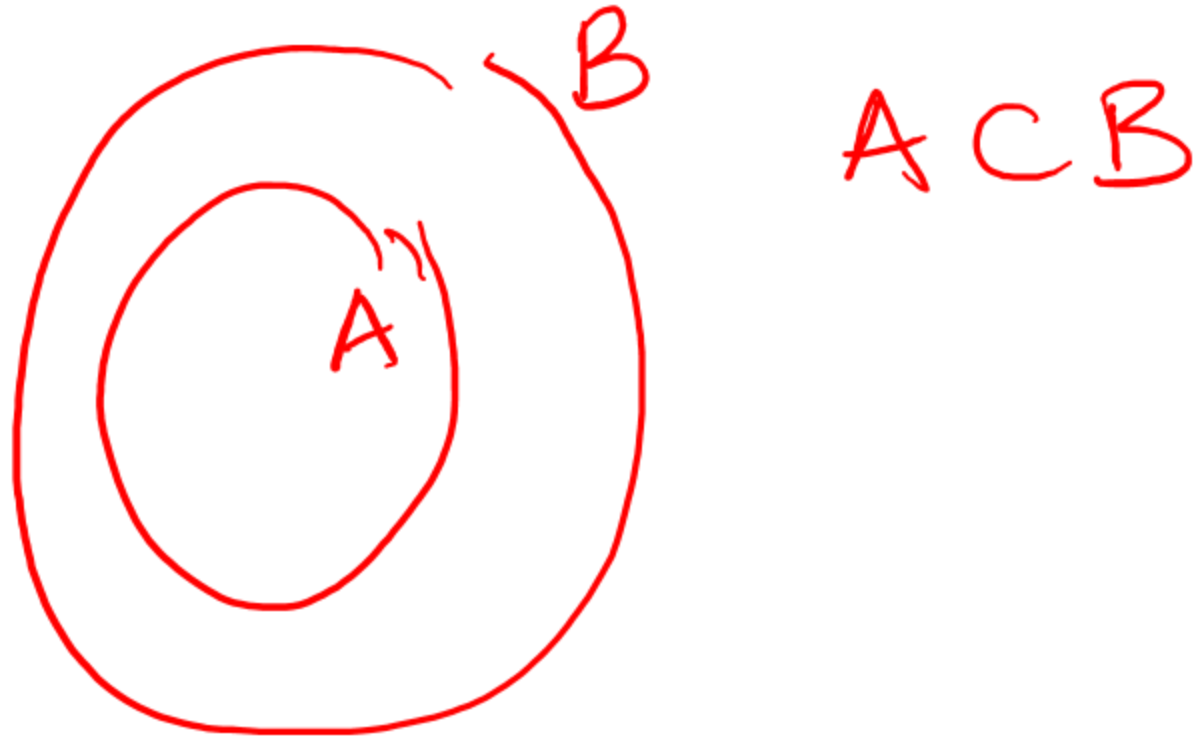
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- Venn diagram:



Sets

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Question

How to prove $A = B$?

- Prove $A \subseteq B$
- Prove $B \subseteq A$

Size of sets

If there are n distinct elements in A , we say that A is a finite set and its **cardinality** is equal to n , denoted by $|A| = n$

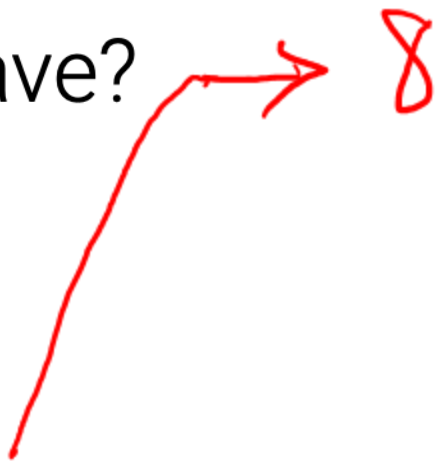
Example:

- $|\emptyset| = 0$
- $|\{1, 2, 3, 4, 5, 6, 7, 8, 9\}| = ?$ 9
- $|\{\underline{1}, \underline{1}, 2, \underline{3}, \underline{3}, 5\}| = ?$ $|\{1, 2, 3, 5\}| = 4$
- $|\{\{1, 2\}, 3\}| = ?$ 2
- $|\{a, b, c, d, \dots, z\}| = ?$ 26
- $|\{x \in \mathbb{Z}^+ \mid x \geq 5\}| = ?$ ∞

Power sets

How many subsets does a finite set have?

Power sets

How many subsets does a finite set have?  8

Example

Let $A = \{1, 2, 3\}$. List all of its subset:

$\emptyset$
 $\{1\}$
 $\{2\}$
 $\{3\}$
 $\{1, 2\}$
 $\{1, 3\}$
 $\{2, 3\}$
 $\{1, 2, 3\}$

Power sets

The set of all subset of A is called **power sets**, denoted by $\mathcal{P}(A)$:

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It can be shown that

$$|\mathcal{P}(A)| = 2^{|A|}.$$

Example:

$$|\mathcal{P}(\{1, 2, 3\})| = 2^3 = 8.$$

Tuples

The ordered n -tuple $(a_1, a_2, \dots, a_n)$ is an ordered collection of n elements with

- a_1 as the first element
- a_2 as the second element
- $\vdots$
- a_n as the last element

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Example

- $(1, 3, 5) \neq (1, 5, 3)$
- Points in $\mathbb{R}^2$: $(1, 5)$ and $(1.2, 3)$
 $\bar{x} \ y \quad x \ y$

Cartesian products

Most discrete structure are based on a particular notion of Cartesian product.

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Definition.

Let A and B be two sets. Then their product, denoted by $A \times B$, is the collection of pairs (a, b) with $a \in A$ and $b \in B$:

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

$$A = \{1, 2, 3\}$$

$$B = \{a, b\}$$

$$(a, b) \neq (b, a)$$

- Note that $A \times B \neq B \times A$.
- We can extend the definition:

$$A \times B = \{(1, a), (2, a), (3, a), (1, b), (2, b), (3, b)\}$$

$$\underline{A_1 \times A_2 \times \cdots \times A_n} = \{(a_1, a_2, \dots, a_n) \mid a_i \in A_i \text{ for } i = 1, 2, \dots, n\}.$$

Cartesian product

$$A \times B \neq B \times A$$

Question.

Is Cartesian product associative: $A \times (B \times C) = (A \times B) \times C$?

Let $a \in A, b \in B, c \in C$

$$\begin{aligned} &A \times (B \times C) \\ &A \times \underbrace{(b, c)} \\ &(a, (b, c)) \end{aligned}$$

$\neq$

$$\begin{aligned} &(A \times B) \times C \\ &(a, b) \times C \\ &((a, b), c) \end{aligned}$$

No.